Effects of Multiple Sclerosis Relapse Treatment during Pregnancy on Offspring Anthropometric Development at Birth and School Age

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1 Introduction

- **To treat MS relapses during pregnancy**, MS guidelines recommend that **high-dose glucocorticoids (GC)** may be used without harm to the fetus.
- Clinical data suggest an **association between prenatal exposure to the GC betamethasone** to promote fetal lung maturation in threatened premature birth and a **reduced postnatal growth**.
- A reduced postnatal growth is a potential risk factor for **metabolic, cardiovascular and neurodevelopmental disease** in later life[1].
- Dose of GC for relapse treatment in MS is up to 50-fold higher compared to obstetric indications.

2 Research Question

Is MS relapse treatment with methylprednisolone (MP) during pregnancy associated with changes in anthropometric outcomes in the offspring at birth and early school age?

3 Methods

- **Cross-sectional multi-center observational study** in 8-13 year old children of mothers with MS prenatally exposed to MS-relapse treatment with MP (exposed group) compared to MP-unexposed children (reference group).
- **Anthropometric measurements** were collected during a scheduled visit, birth data from **medical records** and information on postnatal care using **parental questionnaires.**
- **Insulin-like growth factor (IGF) and its binding protein (IGF-BP)** were determined.

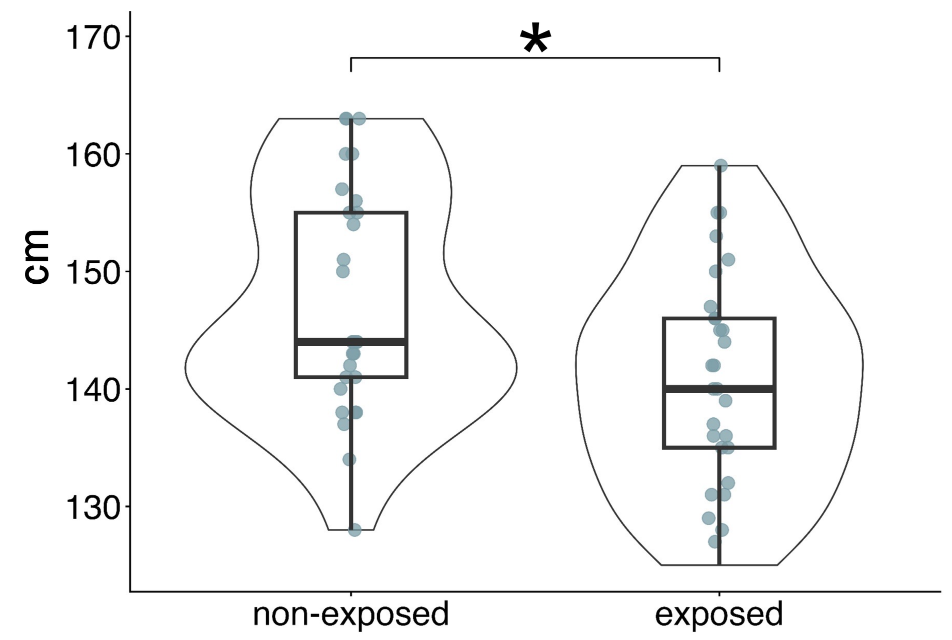
4 Results

Cohort characteristics & birth data

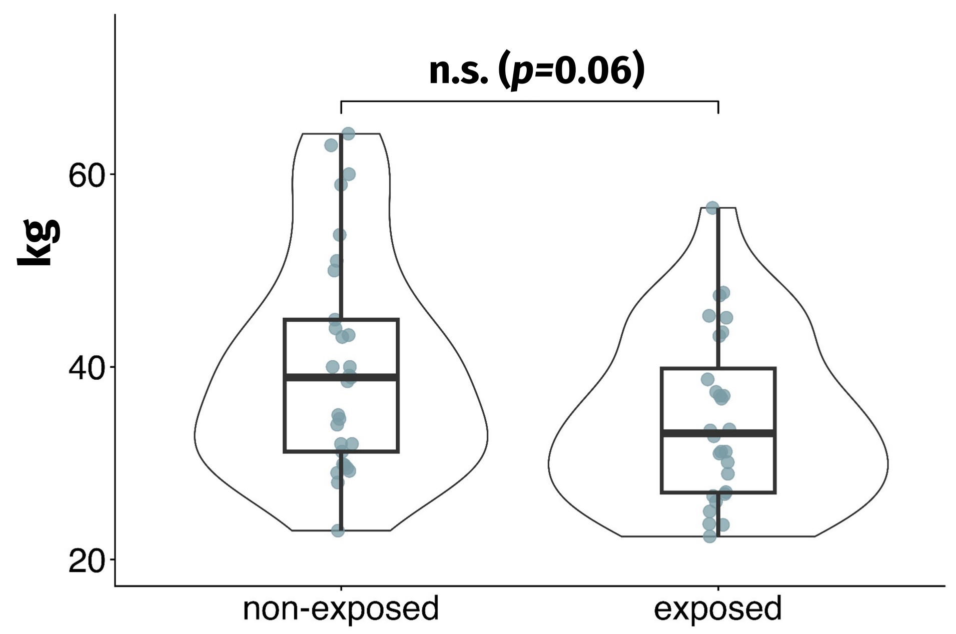
	Reference group (n=29)	MP-group (n=29)	*p*
Child			
Age, years, mean (SD)	10.6 (1.5)	10.1 (1.7)	0.18
Female sex, n (%)	12 (40)	11 (37)	1
Birth weight, g, mean (SD)	3249 (430)	3252 (506)	0.98
Birth length, cm, mean (SD)	50.4 (2.5)	50.6 (2.3)	0.72
Gestational age at birth, weeks, median (IQR)	39.9 (39-41)	38.9 (38– 39)	0.02
Caesarean section, n (%)	10 (33.3)	8 (27.6)	0.78
Breast-fed, n (%)	24 (80)	17 (56.7)	0.10
Parents			
University degree, n (%)	4 (13)	5 (17)	0.79
Maternal height, cm (SD)	170 (7)	167 (5)	0.08
Paternal height, cm (SD)	182 (5)	180 (7)	0.19

- **Duration of pregnancy was reduced by one week in the MP group.**
- **No effect of MP on birth weight and length.**
- Mode of delivery, breastfeeding and parental height did not differ between groups.

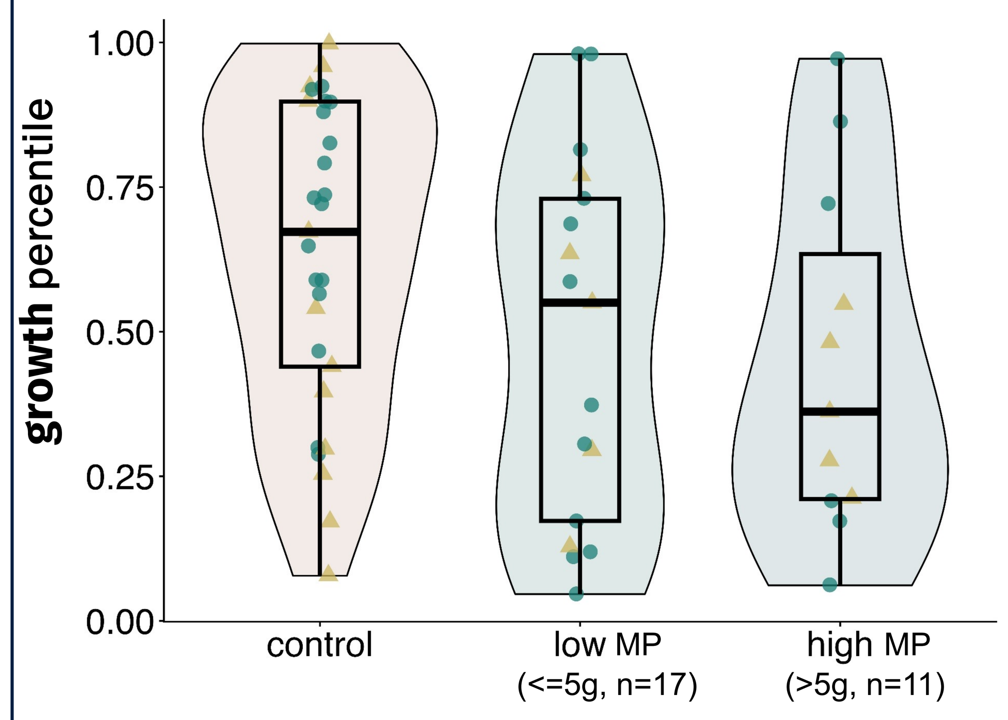
Height at school age



Weight at school age



Dosage effect of MP



- At school age, **height was lower** and weight tended to be lower **in the MP group.**
- Body mass index did not differ between groups.
- Signals for a **dose-dependent effect of MP on height** were found.
- **IGF and IGF-BP** did not differ between groups.

5 Conclusion

- **Treatment of MS relapses with MP during pregnancy** is not associated with fetal growth retardation, but **with reduced growth at school age.**
- This effect may be **dose dependent.**
- As growth **hormones are not affected by MP treatment**, it is important to explore other potential mechanisms such as changes in the **intestinal microbiome.**

The use of MP for MS relapse treatment during pregnancy should be approached with caution and limited to the lowest possible dosage.

References / Funding

[1] Suzuki, K. *J. of dev. origins of health and disease* 9.3 (2018): 266-269.

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